



Barker College

**2005
TRIAL
HIGHER SCHOOL
CERTIFICATE**

Mathematics Extension 1

Staff Involved:

- GIC*
- AES*
- MRB
- VAB
- PJR
- GDH
- RMH

90 copies

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Make sure your Barker Student Number is on ALL pages
- Board-approved calculators may be used
- A table of standard integrals is provided on page 9
- ALL necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged working

PM FRIDAY 12 AUGUST

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value

Total marks – 84

Attempt Questions 1

ALL questions are of equal value

Answer each question on a SEPARATE sheet of paper

Marks

Question 1 (12 marks) [START A NEW PAGE]

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x}{5x}$ 2

(b) Find the acute angle between the lines $2x - y = 0$ and $x + 3y = 0$, giving the answer correct to the nearest minute. 3

(c) Simplify $\frac{e^{\ln(8+3x)}}{64 - 9x^2}$ 2

(d) A is the point $(-2, 1)$ and B is the point (x, y) . The point P $(13, -9)$ divides AB **externally** in the ratio 5 : 3. Find the values of x and y for point B. 2

(e) Solve the inequality $\frac{x}{x-3} \leq 3$ 3

Question 2 (12 marks) [START A NEW PAGE]

(a) Evaluate $\int_0^2 2x\sqrt{1-\frac{x}{2}} dx$ using the substitution $u = 1 - \frac{x}{2}$ 3

(b) Solve for $\theta: \cos\theta = \sin 2\theta$ for $0 \leq \theta \leq 2\pi$ 3

(c) (i) Express $\sin x + \sqrt{3} \cos x$ in the form $R\sin(x + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$ 2

(ii) Hence, sketch $y = \sin x + \sqrt{3} \cos x$ for $-2\pi \leq x \leq 2\pi$ showing any x and y intercepts. 2

(iii) Find the general solution to $\sin x + \sqrt{3} \cos x = \sqrt{2}$ 2

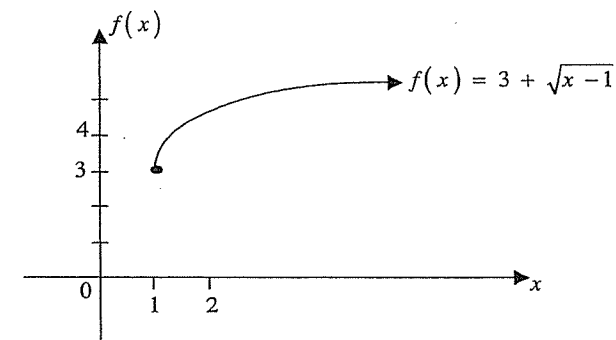
Question 3 (12 marks) [START A NEW PAGE]

(a) Find the domain and range of $y = 3 \sin^{-1}(2x - 1)$ 2
(A sketch of the curve is **not** necessary, but may be helpful).

(b) If $f(x) = (1 + x^2) \tan^{-1} x$ find $f'(x)$ 2

(c) Find $\int \frac{dx}{\sqrt{16 - x^2}}$ 1

(d) The function $f(x) = 3 + \sqrt{x - 1}$ is sketched below.



(i) State the domain of $f(x)$ 1

(ii) Explain why an inverse function, $f^{-1}(x)$, exists. 1

(iii) Find $f^{-1}(x)$ 2

(iv) (α) On what line will the curves $y = f(x)$ and $y = f^{-1}(x)$ intersect? 1

(β) Hence, find the point of intersection of the graphs $y = f(x)$ and $y = f^{-1}(x)$ 2

Question 4 (12 marks) [START A NEW PAGE]

(a) Evaluate $\int_0^{2\pi} \sin^2 2x \, dx$ 2

(b) If $\tan A$ and $\tan B$ are the roots of the equation $3x^2 - 5x - 1 = 0$, find the value of $\tan(A + B)$. 3

(c) Assuming that $r \neq 1$, prove by induction that $a + ar + ar^2 + ar^3 + \dots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1}$ for all positive integers n 3

(d) (i) Given that the curve $y = x \sin^{-1} x$ has only one stationary point, show that this stationary point occurs at $(0, 0)$ and that it is a minimum turning point. 2

(ii) Hence, or otherwise, sketch the curve $y = x \sin^{-1} x$ on the $x - y$ plane. 2

Question 5 (12 marks) [BEGIN A NEW PAGE]

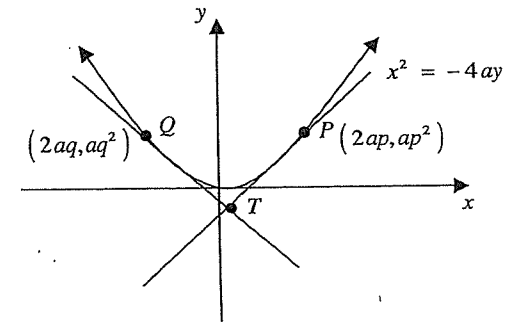
(a) A particle, initially at a fixed point O , is moving in a straight line. After time t seconds, it has displacement x metres from O and its velocity ms^{-1} is given by $v = 6 - 2x$.

(i) Find the acceleration of the particle at the origin. 2

(ii) Show that $t = -\frac{1}{2} \log_e \left(1 - \frac{x}{3} \right)$ and hence find x as a function of t . 3

(iii) What happens to x as t increases without bound? 1

(b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$. The tangents at P and Q meet at T which is always on the parabola $x^2 = -4ay$.



(i) Derive the equations of the tangents to the parabola at P and Q . 2

(ii) Show that T is the point $(a(p + q), apq)$ 2

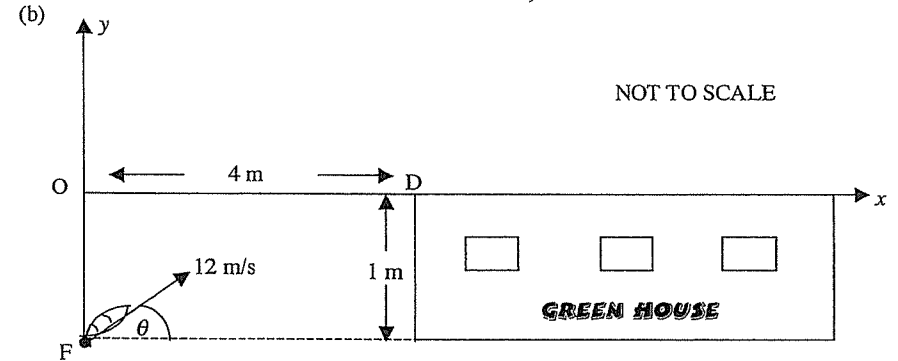
(iii) Show that $p^2 + q^2 = -6pq$ 2

Question 6 (12 marks) [START A NEW PAGE]

- (a) A particle is oscillating in simple harmonic motion such that its displacement x metres from the origin is given by the equation $\frac{d^2x}{dt^2} = -16x$, where t is time in seconds.
- (i) Show that $x = a \cos(4t + \alpha)$ is a solution for the motion of this particle. (a and α are constants) 2
- (ii) Initially, the velocity is 4 ms^{-1} and displacement from the origin is 5 m. Show that the amplitude of the oscillation is $\sqrt{26}$ metres. 2
- (iii) What is the maximum speed of the particle? 2
- (b) In a particular equatorial African swamp, a colony of tsetse flies increases its population (P) according to the differential equation $\frac{dP}{dt} = k(P - 10000)$, where k is the growth rate of the colony. Initially, there were 15 000 tsetse flies and after six months there were 25 000 tsetse flies.
- (i) Show that $P = 10000 + P_0 e^{kt}$ is a solution of the above equation. (k and P_0 are constants) 2
- (ii) Determine P_0 and the growth rate k in exact form. 2
- (iii) Determine the number of tsetse flies after one year. 2

Question 7 (12 marks) [START A NEW PAGE]

- (a) Find the term independent of x in the expansion of $\left(x + \frac{2}{x^2}\right)^{12}$ 3



A football, lying at point F on level ground is 4 metres away from and 1 metre below the top of a flat-roofed long narrow green house. The football is kicked with an initial velocity of 12 m/s at an angle of projection θ .

- (i) Using $g = -10 \text{ ms}^{-2}$, show that the football's trajectory at time t seconds after being kicked may be defined by the equations $x = 12t \cos \theta$ and $y = -5t^2 + 12t \sin \theta - 1$ where x and y are the horizontal and vertical displacements, in metres, of the football from the origin O shown in the diagram. (Neglect air resistance). 3
- (ii) Given that $\theta = 30^\circ$, how far from D will the football land on top of the green house? 3
- (iii) Find the range of values of θ , to the nearest degree, at which the football must be kicked so that it will land to the right of D. 3

End of Paper

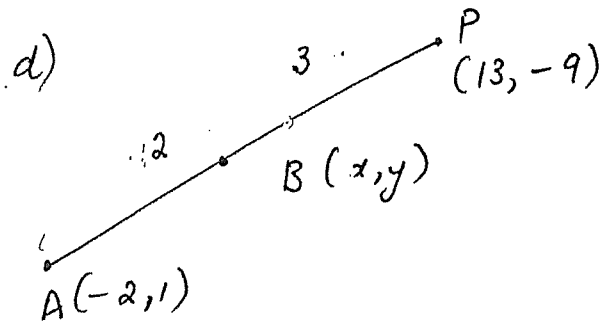
EXT 1 TRIAL 2005

1 a) $\lim_{x \rightarrow 0} \frac{\sin 2x}{5x} = \lim_{2x \rightarrow 0} \frac{\sin 2x}{2x} \times \frac{2}{5}$
 $= 1 \times \frac{2}{5} = \frac{2}{5}$ ✓✓

b) $\tan(\alpha - \beta) = \frac{2 - (-\frac{1}{3})}{1 + 2 \times (-\frac{1}{3})}$ ✓✓✓
 $= \frac{\frac{7}{3}}{\frac{1}{3}} = 7$

acute angle = $81^\circ 52'$ (nearest minute)

c) $\frac{\ln(8+3x)}{64-9x^2} = \frac{8+3x}{(8+3x)(8-3x)}$
 $= \frac{1}{8-3x}, 3x \neq \pm 8$ ✓✓



$x = \frac{2}{5} \times 15 + -2 = 4$

$y = \frac{2}{5}x - 10 + 1 = -3$ ✓✓

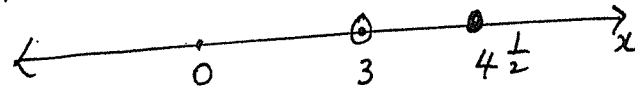
e) $\frac{x}{x-3} \leq 3 \quad x \neq 3$

Solve $x = 3(x-3)$

$\Rightarrow x = 3x - 9$

$\Rightarrow 2x = 9$

$\Rightarrow x = 4\frac{1}{2}$



Test $x < 3$ e.g. $x = 1$

$\frac{1}{1-3} \leq 3$ True $x < 3$

Test $3 < x < 4\frac{1}{2}$ e.g. $x = 4$

$\frac{4}{4-3} \leq 3$ False $x \notin (3, 4\frac{1}{2})$

Test $x > 4\frac{1}{2}$ e.g. $x = 5$

$\frac{5}{5-3} \leq 3$ True $x > 4\frac{1}{2}$

Solution $x < 3$ OR $x \geq 4\frac{1}{2}$ ✓✓✓

2. a) let $u = 1 - \frac{x}{2} \Rightarrow du =$

$\Rightarrow x = 2(1-u) \Rightarrow dx = -2du$

$\Rightarrow$ if $x = 0: u = 1$

if $x = 2: u = 0$

i.e. $\int_0^2 2x \sqrt{1 - \frac{x}{2}} dx = \int_1^0 4(1-u) \sqrt{u} \times -2du$

$= -8 \int_1^0 (\sqrt{u} - u\sqrt{u}) du$

$= 8 \left[\frac{2}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right]_0^1$

$= 8 \left[\frac{2}{3} - \frac{2}{5} - (0 - 0) \right]$

$= 8 \times \frac{10-6}{15} = \frac{32}{15}$ ✓✓✓

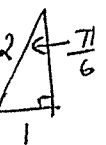
b) $\cos \theta = \sin 2\theta$ for $0 \leq \theta \leq 2\pi$

$\Rightarrow \cos \theta = 2 \sin \theta \cos \theta$

$\Rightarrow \cos \theta (1 - 2 \sin \theta) = 0$

$\Rightarrow \cos \theta = 0$ OR $\sin \theta = \frac{1}{2}$

$\Rightarrow \theta = \frac{\pi}{2}$ OR $\frac{3\pi}{2}$ OR $\frac{\pi}{6}$ OR $\frac{5\pi}{6}$



✓✓✓

$$2c) i) R \sin(x + \alpha)$$

$$= R \sin x \cos \alpha + R \cos x \sin \alpha$$

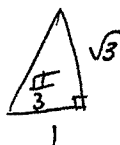
$$= \sin x + \sqrt{3} \cos x$$

$$\Rightarrow R = \sqrt{1^2 + (\sqrt{3})^2} = 2 \quad (R > 0)$$

and

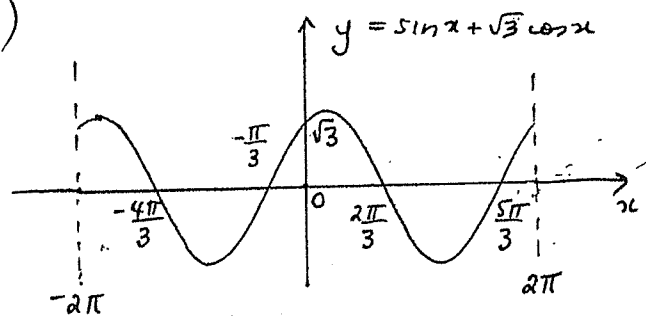
$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{\sqrt{3}}{1} \quad 0 \leq \alpha \leq \frac{\pi}{2}$$

$$\Rightarrow \alpha = \frac{\pi}{3}$$



$$\text{i.e. } \sin x + \sqrt{3} \cos x = 2 \sin\left(x + \frac{\pi}{3}\right)$$

ii)



$$\text{iii) i.e. Solve } 2 \sin\left(x + \frac{\pi}{3}\right) = \sqrt{2}$$

$$\Rightarrow \sin\left(x + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow x + \frac{\pi}{3} = n\pi + (-1)^n \frac{\pi}{4} \quad n \in \mathbb{Z}$$

$$x = n\pi - \frac{\pi}{3} + (-1)^n \frac{\pi}{4}$$

$$2) y = 3 \sin^{-1}(2x - 1)$$

$$\text{Domain: } -1 \leq 2x - 1 \leq 1$$

$$\Rightarrow \{x: 0 \leq x \leq 1\}$$

$$\text{Range: } \{y: -\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}\} \quad \checkmark \checkmark$$

$$b) f(x) = (1 + x^2) \tan^{-1} x$$

$$f'(x) = 2x \tan^{-1} x + \frac{(1 + x^2)}{1 + x^2}$$

$$= 1 + 2x \tan^{-1} x \quad \checkmark \checkmark$$

$$c) \int \frac{dx}{\sqrt{16 - x^2}} = \sin^{-1} \frac{x}{4} + C, \quad -4 < x < 4 \quad \checkmark$$

$$d) i) \text{ Domain} = \{x: x \geq 1\} \quad \checkmark$$

$$ii) f(x) \text{ is one to one} \quad \checkmark$$

$$iii) \text{ Let } y = f^{-1}(x) \text{ then}$$

$$x = 3 + \sqrt{y - 1} \quad y \geq 1, x \geq 3$$

gives $f^{-1}(x)$.

$$\text{i.e. } (x - 3)^2 = y - 1$$

$$\Rightarrow y = (x - 3)^2 + 1 \quad x \geq 3$$

$$\text{i.e. } f^{-1}(x) = x^2 - 6x + 10 \quad \checkmark \checkmark$$

$$3d) iv) \alpha) y = x$$

$\beta)$ Hence graphs intersect where

$$x = (x - 3)^2 + 1$$

$$\Rightarrow x = x^2 - 6x + 9 + 1$$

$$\Rightarrow x^2 - 7x + 10 = 0$$

$$\Rightarrow (x - 5)(x - 2) = 0$$

$$\Rightarrow x = 5 \text{ or } 2 \text{ but } x \geq 3$$

Point of intersection:

$$(5, 5)$$

$$4a) \int_0^{2\pi} \sin^2 2x \, dx$$

Now

$$\cos 2A = 1 - 2\sin^2 A$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$= \int_0^{2\pi} \frac{1 - \cos 4x}{2} \, dx$$

$$= \frac{1}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{2\pi}$$

$$= \frac{1}{2} [2\pi - 0 - (0 - 0)] = \pi \quad \checkmark \checkmark$$

$$b) \tan A + \tan B = \alpha + \beta = \frac{5}{3}$$

$$\tan A \times \tan B = \alpha \beta = -\frac{1}{3}$$

$$\tan(A + B) = \frac{\frac{5}{3}}{1 + \frac{1}{3}} = \frac{5}{3} \div \frac{4}{3}$$

$$= \frac{5}{4}$$

$\checkmark \checkmark \checkmark$

c) Let $S_n = a + ar + ar^2 + ar^3 + \dots + ar^{n-1}$
 Show $S_n = \frac{a(r^n - 1)}{r - 1} \quad n \in \mathbb{Z}^+$

Step 1 $n=1$
 LHS: $S_1 = a$ RHS: $\frac{a(r^1 - 1)}{r - 1} = \frac{a(r - 1)}{r - 1} = a = S_1$

as required

Step 2: Assume true for $n=k$

Show true for $n=k+1$

i.e. show that $S_{k+1} = \frac{a(r^{k+1} - 1)}{r - 1}$

Now $S_{k+1} = S_k + T_k$

$= \frac{a(r^k - 1)}{r - 1} + ar^k$ by assumption

$= \frac{a(r^k - 1)}{r - 1} + ar^k \frac{(r - 1)}{r - 1}$
 $= \frac{ar^k - a + ar^{k+1} - ar^k}{(r - 1)}$

$= \frac{-a + ar^{k+1}}{r - 1}$

$= \frac{a(r^{k+1} - 1)}{r - 1}$ as required

Step 3 Show true for all n :

Step 1 showed statement true for $n=1$.

By Step 2 if true for $n=1$ Then true for $n=1+1=2$. Similarly if true for $n=2$ then true for $n=2+1=3$ and so on.

i.e. Statement is true for all $n=1, 2, 3, \dots$ all $n \in \mathbb{Z}^+$.

4 d) i) $y = x \sin^{-1} x$

$\frac{dy}{dx} = \sin^{-1} x + \frac{x}{\sqrt{1-x^2}}$

Stationary points occur when $\frac{dy}{dx} = 0$

i.e. when $\sin^{-1} x + \frac{x}{\sqrt{1-x^2}} = 0$

Let $x=0 \Rightarrow \frac{dy}{dx} = \sin^{-1} 0 + \frac{0}{\sqrt{1-0}} = 0 + 0 = 0$

hence $x=0$ is a stationary pt.

Test nature find sign of $\frac{dy}{dx}$

for $x = 0 - \epsilon$ and $x = 0 + \epsilon$

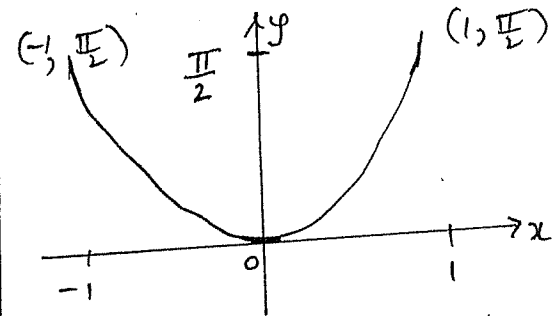
x	-0.5	0	$x=0.5$
$\frac{dy}{dx}$	$-\frac{\pi}{6} - \frac{1}{\sqrt{3}}$	0.0	$\frac{\pi}{6} + \frac{1}{\sqrt{3}}$
$\frac{dy}{dx}$	-1.1	0.0	+1.1
	decreasing	stationary	increasing

i.e. minimum at $x=0, y=0$ (0,0)
 (no other S.P. in domain $-1 \leq x \leq 1$) ✓

d) ii) Domain $-1 \leq x \leq 1$

Range $0 \leq y \leq \frac{\pi}{2}$

S.P. a min at $x=0, y=0$



$\frac{dy}{dx} \rightarrow -\infty$ as $x \rightarrow -1^+$
 $\frac{dy}{dx} \rightarrow +\infty$ as $x \rightarrow +1^-$ ✓

5a) $t=0, x=0 \quad v = 6 - 2x$

i) $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$= \frac{d}{dx} \frac{(6-2x)^2}{2} = \frac{d}{dx} \frac{36 - 24x + 4x^2}{2}$

$= -12 + 4x$

$= -12$ when $x=0$ ✓

ii) $\frac{dx}{dt} = v = 6 - 2x$

$\Rightarrow \frac{dt}{dx} = \frac{1}{6-2x}$

$\Rightarrow t = -\frac{1}{2} \ln(6-2x) + C$

5a) ii) ctd

when $x=0, t=0$ i.e.

$$0 = -\frac{1}{2} \ln 6 + C$$

$$\Rightarrow t = -\frac{1}{2} \ln(6-2x) + \frac{1}{2} \ln 6$$

$$= -\frac{1}{2} \ln \frac{6-2x}{6}$$

$$= -\frac{1}{2} \ln \left(1 - \frac{x}{3}\right)$$

$$\Rightarrow \ln \left(1 - \frac{x}{3}\right) = -2t$$

$$\Rightarrow 1 - \frac{x}{3} = e^{-2t}$$

$$\Rightarrow \frac{x}{3} = 1 - e^{-2t}$$

$$\Rightarrow x = 3 - 3e^{-2t}$$

iii) $x \rightarrow 3$

$$b) i) \frac{dy}{dx} = \frac{dy/dx}{dp/dp}$$

$$TP = 2ap/2a = p$$

$$\frac{dy}{dx}_{TP} = 2aq/2a = q$$

$$\text{tangent TP: } y - ap^2 = p(x - 2ap)$$

$$\Rightarrow y - ap^2 = px - 2ap^2$$

$$\Rightarrow y = px - ap^2$$

similarly tangent TP

$$\Rightarrow y = qx - aq^2$$

ii) Solve TP, TP simultaneously to find T

$$\text{i.e. } px - ap^2 = qx - aq^2$$

$$\Rightarrow (p-q)x = a(p^2 - q^2)$$

$$\Rightarrow x = a \frac{(p+q)(p-q)}{p-q} \quad (p \neq q)$$

$$\Rightarrow x = a(p+q)$$

$$\therefore y = pa(p+q) - ap^2 \\ = ap^2 + apq - ap^2 \\ = apq$$

i.e. T is $(a(p+q), apq)$ ✓

iii) Now T lies on $x^2 = -4ay$

$$\therefore [a(p+q)]^2 = -4a(apq)$$

$$\Rightarrow a^2(p+q)^2 = -4a^2pq$$

$$\Rightarrow (p+q)^2 = -4pq \quad (a \neq 0)$$

$$\Rightarrow p^2 + 2pq + q^2 = -4pq$$

$$\Rightarrow p^2 + q^2 = -6pq \quad \checkmark$$

6.i) Show that $\frac{d^2x}{dt^2} = -16x$

$$x = a \cos(4t + \alpha)$$

$$\Rightarrow \frac{dx}{dt} = -4a \sin(4t + \alpha)$$

$$\Rightarrow \frac{d^2x}{dt^2} = -16a \sin(4t + \alpha)$$

$$= -16x \quad \text{as required} \quad \checkmark$$

ii) $t=0, v=4$

$$\Rightarrow 4 = -4a \sin \alpha \Rightarrow \sin \alpha = -\frac{1}{a}$$

$t=0, x=5$

$$\Rightarrow 5 = a \cos \alpha \Rightarrow \cos \alpha = \frac{5}{a}$$

$$\Rightarrow \left(-\frac{1}{a}\right)^2 + \left(\frac{5}{a}\right)^2 = 1$$

$$\Rightarrow 1 + 25 = a^2$$

$$\Rightarrow a = \sqrt{26} \quad \checkmark \checkmark$$

iii) Max speed when $|\sin(4t + \alpha)| = 1$

$$\text{i.e. } |v|_{\max} = |-4 \times \sqrt{26} \times 1|$$

$$= 4\sqrt{26}$$

b) i) $P = 10000 + P_0 e^{kt} \Rightarrow P_0 e^{kt} = P - 10000$
and

$$\Rightarrow \frac{dP}{dt} = k P_0 e^{kt} \\ = k(P - 10000) \quad \text{as required} \quad \checkmark$$

ii) $t=0, P=15000$ — ①

$t=6, P=25000$ — ②

$$\text{①} \Rightarrow 15000 = 10000 + P_0 e^{k \times 0}$$

$$\Rightarrow P_0 = 5000$$

$$\text{i.e. } P = 10000 + 5000 e^{kt}$$

$$\text{②} \Rightarrow 25000 = 10000 + 5000 e^{6k}$$

$$\Rightarrow \frac{15000}{5000} = e^{6k}$$

$$\Rightarrow 6k = \ln 3$$

$$\Rightarrow k = \frac{\ln 3}{6} \quad \checkmark \checkmark$$

6b) iii) $t = 12$

$$\Rightarrow p = 10000 + 5000e^{\frac{\ln 3 \times 2}{6}} \\ = 10000 + 5000e^{(\ln 3) \times 2} \\ = 10000 + 5000 \times 3^2 \\ = \underline{55000} \text{ flies after 1 year.}$$

$$7a) T_{r+1} = \binom{12}{r} x^{12-r} \left(\frac{2}{x^2}\right)^r \\ = \binom{12}{r} x^{12-r} x^{-2r} \\ = \binom{12}{r} x^{12-3r}$$

For term independent of x
require x^0 , i.e. $12-3r=0$
 $r=4$

5th term is $\binom{12}{4} x^4$

$$1. \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} \times 16$$

$= 7920$

7b)

i) at $t=0$, $x=0$, $y=-1$

$$\dot{x} = 12 \cos \theta, \dot{y} = 12 \sin \theta$$

given $\ddot{x}(t)=0$, $\ddot{y}(t)=-10$

$$\dot{x} = \int \ddot{x}(t) dt$$

$$= 0 + c$$

$$\Rightarrow c = 12 \cos \theta$$

$$\Rightarrow \dot{x} = 12 \cos \theta$$

$$\Rightarrow x = \int \dot{x} dt$$

$$= 12t \cos \theta + D$$

$$D=0$$

$$\Rightarrow x = 12t \cos \theta$$

$$\dot{y} = \int \ddot{y}(t) dt$$

$$= -10t + E$$

$$E = 12 \sin \theta$$

$$\Rightarrow \dot{y} = -10t + 12 \sin \theta$$

$$\Rightarrow y = \int (-10t + 12 \sin \theta) dt$$

$$= -5t^2 + 12t \sin \theta + F$$

$$\Rightarrow -1 = F$$

$$\Rightarrow y = -5t^2 + 12t \sin \theta - 1$$

ii) Hits roof when $y=0$

$$1.e. y=0 = -5t^2 + 12t \sin 30^\circ - 1$$

$$\Rightarrow -5t^2 + 6t - 1 = 0$$

$$\Rightarrow 5t^2 - 6t + 1 = 0$$

$$\Rightarrow (5t-1)(t-1) = 0$$

If ball hits roof above so on downward trajectory i.e. at greater value of t : $t=1$

$$1.e. x = 12 \times 1 \times \cos 30^\circ$$

$$= 6\sqrt{3} \text{ which is } (6\sqrt{3}-4)m$$

from D

iii) To land to right of D without hitting greenhouse, ball must cross OD i.e. $x < 4$ for $-1 < y < 0$.

Limit is through D(4,0):

$$x: 12t \cos \theta = 4 \Rightarrow t = \frac{4}{12 \cos \theta}$$

$$y: -5t^2 + 12t \sin \theta - 1 = 0 \quad \text{in } y$$

$$\Rightarrow -5\left(\frac{1}{3 \cos \theta}\right)^2 + 12\left(\frac{1}{3 \cos \theta}\right) \sin \theta - 1 = 0$$

$$\Rightarrow -\frac{5}{9} \sec^2 \theta + 4 \tan \theta - 1 = 0$$

$$\Rightarrow -5(1 + \tan^2 \theta) + 36 \tan \theta - 9 = 0$$

$$\Rightarrow -5 \tan^2 \theta + 36 \tan \theta - 14 = 0$$

$$\Rightarrow \tan \theta = \frac{36 \pm \sqrt{1016}}{10}$$

$$\theta > 22.48^\circ \text{ and } < 81.62^\circ \text{ (2dp)}$$

20.010 / 0.20 min rest degree